

Series Convergence — Geometric, p -Series, and the n th-Term Test

Answer Key

AP Calculus BC

Quick Answers

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|------------------|------------------|-----------|
| 1. 5 | 4. $\frac{9}{5}$ | 7. 72, 84 |
| 2. $\frac{1}{2}$ | 5. $\frac{3}{5}$ | 8. — |
| 3. — | 6. —, 0 | |
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Curriculum

Level: AP Calculus BC

LIM-7, LIM-8 – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–5 of 5 across 10 tagged checks.

How to grade this sheet: on the AP exam these questions are scored on the *justification*, not the verdict. A correct “diverges” with no named test and no parameter earns nothing. Look for three things in every answer: the family (geometric, p -series, neither), the parameter that decides it (r or p) with its comparison to 1, and the conclusion stated as a consequence of that comparison. The one place students lose the most is problem 6 and problem 8 — the belief that terms shrinking to zero is enough.

1. A geometric series, read off directly.

(a) Write the series with its index starting where the problem starts it, $n = 0$: the first term is the $n = 0$ term, $a = 3 \left(\frac{2}{5}\right)^0 = 3$, and the ratio of consecutive terms is $r = \frac{2}{5}$.

(b) It is geometric with $|r| = \frac{2}{5} < 1$, so the geometric series test applies and the series converges.

(c) Use $S = \frac{a}{1-r}$ with those two values:

$$S = \frac{3}{1 - \frac{2}{5}} = \frac{3}{\frac{3}{5}} = 3 \cdot \frac{5}{3}$$

So the sum is $\boxed{5}$.

Check it against Model A: the running totals on the front page are 3, 4.2, 4.68, 4.872, 4.9488 — climbing toward 5 and never passing it. A student whose formula gives a number smaller than the first term has inverted the fraction.

2. The n th-term test, stated correctly.

(a) Divide numerator and denominator by the highest power of n present, here n itself:

$$\lim_{n \rightarrow \infty} \frac{n}{2n+5} = \lim_{n \rightarrow \infty} \frac{1}{2 + \frac{5}{n}} = \frac{1}{2+0}$$

So the limit is $\boxed{1/2}$.

(b) The n th-term test (test for divergence) says: if $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum a_n$ diverges. Here the limit is $\frac{1}{2}$, which is not zero, so $\sum_{n=1}^{\infty} \frac{n}{2n+5}$ **diverges**.

The reason, in one sentence: a sum can only settle down if the pieces being added eventually become negligible; these terms stay near $\frac{1}{2}$ forever, so each new term adds about another half and the partial sums grow without bound.

3. A p -series, classified by its exponent.

(a) $\sum \frac{1}{n^3}$ has the form $\sum \frac{1}{n^p}$ with $p = 3$.

(b) The comparison that decides every p -series is p against 1:

$$\boxed{3 > 1}$$

(c) The p -series test says $\sum 1/n^p$ converges when $p > 1$ and diverges when $p \leq 1$. Since $3 > 1$, this series **converges**.

Note for the student who asks: the test gives no value for the sum, and there is no elementary formula for it. Convergence and evaluation are different questions — geometric series are unusual in answering both.

4. A geometric series with a negative ratio.

(a) With the index starting at $n = 1$, the exponent $n - 1$ is 0 in the first term, so $a = 6$ and $r = -\frac{1}{3}$. A negative ratio makes the terms **alternate** in sign: $6, -2, \frac{2}{3}, -\frac{2}{9}, \dots$

(b) What matters for convergence is the size of r , not its sign: $|r| = \frac{1}{3} < 1$, so the series converges. The same formula applies, with the sign carried through:

$$S = \frac{6}{1 - (-\frac{1}{3})} = \frac{6}{\frac{4}{3}} = 6 \cdot \frac{3}{4}$$

The sum is $\boxed{9/2}$.

Watch for: using $1 - \frac{1}{3}$ in the denominator, which gives 9. The formula subtracts r , and r is negative here, so the denominator gets larger, not smaller — and the sum must be smaller than the first term 6, which 9 is not.

5. The n th-term test on a ratio of quadratics.

(a) Divide through by n^2 , the highest power present:

$$\lim_{n \rightarrow \infty} \frac{3n^2 + 1}{5n^2 - n} = \lim_{n \rightarrow \infty} \frac{3 + \frac{1}{n^2}}{5 - \frac{1}{n}} = \frac{3 + 0}{5 - 0}$$

The limit is $\boxed{3/5}$.

(b) The limit is not zero, so by the n th-term test the series **diverges**.

(c) The classmate's rule is wrong because it ignores the *degrees*. A rational a_n tends to zero only when the denominator's degree is strictly higher than the numerator's. Here both are degree 2, so the terms settle at the ratio of the leading coefficients, $\frac{3}{5}$, and never shrink at all.

Watch for: answering 0 here. That is the single most common error on this problem, and it inverts the conclusion — it would make the n th-term test say nothing when in fact the test settles the question outright.

6. Where the n th-term test runs out and the p -series test does not.

(a) $\frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$, so $p = \frac{1}{2}$, and the comparison is

$$\boxed{1/2 < 1}$$

(b) As n grows, $\sqrt{n}$ grows without bound, so its reciprocal shrinks to nothing:

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = \boxed{0}$$

(c) *Inconclusive* means the test has been applied correctly and simply returns no information. The n th-term test has one direction only: a nonzero limit proves divergence; a zero limit proves **nothing**, and the series may do either. So the work in (b) is not wasted, but it does not answer the question.

The p -series test does answer it: $p = \frac{1}{2} \leq 1$, so $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ **diverges**.

This pairing is the point of the sheet. Two tests, applied to the same series, give “no information” and “divergent”. Neither contradicts the other, because “inconclusive” is not a verdict.

7. A bouncing ball as a geometric series.

(a) **Where the 24 comes from:** the ball rises to $12\left(\frac{3}{4}\right)$ after the first bounce and then falls the same distance, so that bounce contributes $2 \cdot 12 \cdot \frac{3}{4}$. Every later bounce does the same thing, so the n th bounce contributes $2 \cdot 12\left(\frac{3}{4}\right)^n = 24\left(\frac{3}{4}\right)^n$ — the 24 is twice the drop height, because each bounce is counted up and down.

This is geometric with first term ($n = 1$) equal to $24 \cdot \frac{3}{4} = 18$ and ratio $\frac{3}{4}$:

$$\sum_{n=1}^{\infty} 24\left(\frac{3}{4}\right)^n = \frac{18}{1 - \frac{3}{4}} = \frac{18}{\frac{1}{4}}$$

The bouncing accounts for 72 ft.

(b) The original drop is not part of that sum — it happens before the first bounce — so add it:

$$12 + 72$$

The total vertical distance is 84 ft.

The idea worth stating out loud: the ball bounces infinitely many times and travels a finite distance. Infinitely many nonzero pieces can have a finite total, which is exactly what a convergent series means. (It also stops in finite time, for the same reason applied to the durations.)

Watch for: using $a = 24$ instead of $a = 18$. The index starts at $n = 1$, so the first term already carries one factor of $\frac{3}{4}$; $a = 24$ gives 96 and a total of 108.

8. Challenge: what the n th-term test can and cannot do. (Open response — read the reasoning.)

A full-credit paragraph makes all three moves.

(i) **The test is one-directional.** The n th-term test says: if $\lim_{n \rightarrow \infty} a_n \neq 0$ then $\sum a_n$ diverges. Its converse is not a theorem. $\lim a_n = 0$ is a *necessary* condition for convergence, never a sufficient one, so from $\lim \frac{1}{n} = 0$ nothing whatever follows. The classmate has used the test backwards.

(ii) **The correct classification.** $\sum_{n=1}^{\infty} \frac{1}{n}$ is the harmonic series — a p -series with $p = 1$. The p -series test requires $p > 1$ for convergence, and $1 > 1$ is false, so the harmonic series **diverges**. Model B on the front page shows it: the running totals reach only about 7.485 after a thousand terms, but they keep climbing, and they pass every bound eventually (each block of terms from $2^k + 1$ to 2^{k+1} adds more than $\frac{1}{2}$).

(iii) **The contrast.** $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ has terms that also go to zero, and it converges, to 1. So two series with the same limiting behaviour of the terms land on opposite sides of the question. What separates them is *how fast* the terms shrink: geometrically fast is enough, while $\frac{1}{n}$ shrinks too slowly to stop the total from growing.

Accept any correct convergent geometric series in (iii). Do not accept a paragraph that asserts the harmonic series diverges without naming $p = 1$ or otherwise giving a reason; the verdict alone is what the classmate could have guessed.

Open explanation — open response.